

Day 10 — Final Share and Assessment

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: present design + summative quiz.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. ““auspicious arithmetic” — the Indian term for magic squares”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Design share (10 min).** Each student shows their magic-square art. Brief explanation of the math.
2. **Summative quiz (25 min).** Magic constant formula, stair-step algorithm, historical context. See `quizzes/summative.md`.
3. **Critical-thinking close (5 min).** “*What’s the most enduring lesson from this module? Is it the math (algorithm), the history (Nārāyaṇa, Khajuraho), or the cross-cultural insight (magic squares everywhere)?*” Each student writes 2 sentences.
4. **Linkage.** Module 15 (Capstone) — magic-square art is one of the project options.
5. **Module close.** “*Designing your own magic square required CHOOSING constraints. What other constrained-design problems are like this?*” (Sudoku, crosswords, music composition, architecture.) Open discussion.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Final Share and Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).